



45 – 65 hrs

V2.0 (MUST HAVE)

40 – 60 hrs

V3.0 (GOOD TO HAVE)

40 – 65 hrs

V4.0+ (NICE TO HAVE)

125 – 190 hrs

TOTAL LIFECYCLE EFFORT

🌟 New Architecture Enhancements Included in Roadmap

- **Modular Monolith:** Unified FastAPI codebase cleanly divided into app/services/intake, audio, guardrails, and pdf.
- **Production Observability:** Langfuse (LLM cost & prompt tracing), Sentry (real-time error stack tracking), Slowapi (rate limiting).

🚀 Version 2.0 Roadmap — "The Foundation & Conversational Upgrade" (Must Have)

MODULE	TECHNICAL SCOPE & ENHANCEMENTS	PRIORITY	EST. HOURS
Modular Monolith Stack	Decouple to React (Vite) frontend on Vercel + FastAPI unified backend on Render. Modular folder architecture ready for future microservices split.	MUST HAVE	10 – 15 hrs
Slot-Filling Extraction	OpenAI Structured Outputs (Pydantic v2, strict=True) extracting multi-field patient input in 1 turn; asks only for missing slots.	MUST HAVE	8 – 12 hrs
Full Voice Pipeline	Browser audio (MediaRecorder) → Whisper STT → Chat Engine → tts-1 audio output stream for hands-free intake.	MUST HAVE	15 – 20 hrs
3-Layer Guardrails	Layer 1: Emergency red-flag regex. Layer 2: LLM intent classifier. Layer 3: Redirect pivot engine to eliminate intent drift.	MUST HAVE	8 – 12 hrs
Observability & Tokens	Integration of Langfuse for prompt/cost tracing, Sentry for crash reporting, and live token usage tracking in admin UI.	MUST HAVE	4 – 6 hrs

📱 Version 3.0 Roadmap — "Accessibility, Reach, & Speed" (Good to Have)

MODULE	TECHNICAL SCOPE	PRIORITY	EST. HOURS
WhatsApp Bot	Twilio / WhatsApp API integration for voice-note based intake directly in WhatsApp with auto PDF delivery.	GOOD TO HAVE	15 – 20 hrs
Multilingual Engine	Sarvam AI / Indic speech integrations mapping local dialects and idioms directly to standard medical English.	GOOD TO HAVE	12 – 18 hrs
Async & Redis Cache	FastAPI `asyncio` refactor + Upstash Redis caching for prompts/translations and IP rate limiting via Slowapi.	PERFORMANCE	8 – 12 hrs
Caregiver Mode	1-tap UI toggle allowing family members to assist elderly relatives, shifting prompts to 3rd person ("his/her symptoms").	GOOD TO HAVE	5 – 10 hrs

🏥 Version 4.0+ Roadmap — "The Clinical Powerhouse" (Nice to Have)

MODULE	TECHNICAL SCOPE	PRIORITY	EST. HOURS
Visual Body Map	Interactive 2D/3D human body SVG in React; taps translate directly to medical anatomical terms.	NICE TO HAVE	10 – 15 hrs
RxNorm Drug API	NIH RxNorm API search integration for current medications with automated allergy warnings.	NICE TO HAVE	8 – 12 hrs
SMART-on-FHIR Export	HL7 FHIR `QuestionnaireResponse` JSON exporter for direct ingestion into hospital EHR systems (Epic/Cerner).	NICE TO HAVE	10 – 15 hrs
Post-Visit Decoder	Consultation audio uploader converting doctor's instructions into plain-language home care action plans.	NICE TO HAVE	8 – 12 hrs